

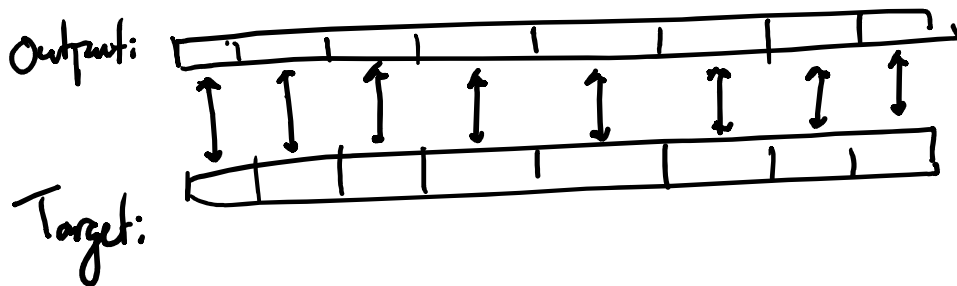
Goal: $\frac{\partial L}{\partial w_i}$

Learn parameters in recurrence function.

Consider Vanilla RNN with
 Scalar input, hidden state, output
 Many-to-many design pattern

Compute loss on sequence of outputs:

Assume: $L = \sum_t L_t$



Recurrence equations for Vanilla Unit:

$$a_t = b + W h_{t-1} + U x_t$$

$$h_t = \tanh(a_t)$$

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$$o_t = c + Vh_t$$

Preliminaries:

$$\frac{\partial L}{\partial o_t} = \frac{\partial L_t}{\partial o_t}$$

$$\frac{\partial L}{\partial h_t} = \frac{\partial L}{\partial o_t} \frac{\partial o_t}{\partial h_t} + \frac{\partial L}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial h_t}$$

$$= \frac{\partial L_t}{\partial o_t} V + \frac{\partial L}{\partial h_{t+1}} \frac{\partial h_{t+1}}{\partial a_{t+1}} \frac{\partial a_{t+1}}{\partial h_t}$$

$$= \frac{\partial L_t}{\partial o_t} V + \frac{\partial L}{\partial h_{t+1}} \tanh'(a_{t+1}) W$$

$$\frac{\partial L}{\partial h_T} = \frac{\partial L_T}{\partial o_T} V$$

$$\frac{\partial L}{\partial h_{T-1}} = \frac{\partial L_{T-1}}{\partial o_{T-1}} V + \frac{\partial L}{\partial h_T} \tanh'(a_T) W$$

Compute change in loss for parameters

$$\frac{\partial L}{\partial c} = \sum_t \frac{\partial L}{\partial o_t} \frac{\partial o_t}{\partial c}$$

$$= \sum_t \frac{\partial L_t}{\partial o_t}$$

$$\frac{\partial L}{\partial v} = \sum_t \frac{\partial L}{\partial o_t} \frac{\partial o_t}{\partial v}$$

$$= \sum_t \frac{\partial L_t}{\partial o_t} h_t$$

$$\frac{\partial L}{\partial b} = \sum_t \frac{\partial L}{\partial h_t} \frac{\partial h_t}{\partial a_t} \frac{\partial a_t}{\partial b}$$

$$= \sum_t \frac{\partial L}{\partial h_t} \tanh'(a_t)$$

$$\frac{\partial L}{\partial w} = \sum_t \frac{\partial L}{\partial h_t} \frac{\partial h_t}{\partial a_t} \frac{\partial a_t}{\partial w}$$

$$= \sum_t \frac{\partial L}{\partial h_t} \tanh'(a_t) h_{t-1}$$

$$\begin{aligned}\frac{\partial L}{\partial u} &= \sum_t \frac{\partial L}{\partial h_t} \frac{\partial h_t}{\partial a_t} \frac{\partial a_t}{\partial u} \\ &= \sum_t \frac{\partial L}{\partial h_t} \tanh'(a_t) x_t\end{aligned}$$

Computing $\partial L / \partial w_i$ on RNNs:

1. Forward propagate input x_t to compute a_t, h_t, o_t for all times t .
2. Backpropagate over time to compute $\partial L / \partial h_t$ for all t .
3. Compute partial derivatives of loss wrt. parameters using $\partial L / \partial h_t$ computed in Step 2